

Sumit Kumar

5th Year Undergraduate

Department of Computer Science and Engineering

Indian Institute of Technology Kharagpur

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in Sumit Kumar SumitKumar-17

Academic Qualifications

Year	Degree/Certificate	Institute	CGPA/%
2022 - 2027	5 year Dual Degree Course(BTech. + MTech.) in Computer Science	Indian Institute of Technology, Kharagpur	8.82/10
2022	Class XII (CBSE)	Delhi Public School, Kalyanpur	94.20%
2020	Class X (CBSE)	Delhi Public School, Kalyanpur	96.60%

Work Experiences

Software Engineer Intern, GitHub Actions - Ubicloud

Aug'26-Sep'26

- Split **GitHub Runner vCPU quotas** by provider, capped **AWS spillover** per customer, and added monthly spending limits
- Modeled **runner-pool demand** with an **EWMA** estimator (**Little's Law**), raising simulated hit ratio from **36% to 94%**
- Fixed **GitHub webhook redelivery** misses under high volume using a persisted, time-bounded cursor replacing page scans
- Added **host exclusions**, **AWS AMI** cross-account sharing, a runner-listing **CLI**, and **Ubuntu 26.04** CI test support

Software Engineer Intern - Atlassian

May'26-Jul'26

- Shipped **Bulk Archive** and **Bulk Delete Teams** across all Atlassian orgs via **GraphQL mutations** and frontend dialogs
- Replaced a **3-4s manual deletion flow** with one API call using hierarchical async deletion on **AWS SQS** and **DynamoDB**
- Proposed a fix for **notification spam** in Bulk Archive/Delete and validated it via **Splunk** in staging and production
- Instrumented API latency, failure, and counter metrics with **SignalFx** dashboards and **Terraform**-managed alert rules

Backend Engineer Intern - Steps AI

Jun'25-May'26

- Engineered async **Python microservices** with **FastAPI** and **Pydantic**, serving agent, copilot, voice, and FAQ APIs
- Orchestrated **LangGraph** and **LangChain** agents with streaming, checkpointing, and **Milvus/Cohere** retrieval
- Built a **DAG-based workflow platform** on **PostgreSQL/Redis** with retries, conditional steps, and connector actions
- Automated **Shopify/WooCommerce** product sync and embedding ingestion via **Temporal** workflows across **S3** and **Milvus**
- Deployed services with **Docker** and **ArgoCD** GitOps, adding **Sentry/OpenTelemetry** and secure **OAuth** refresh

System Architecture and Platform Design - Spring Fest

Jul'24-Apr'25

- Scaled backend for **100M+ API requests** by deploying with **Nginx**, **Redis caching** on **4 AWS EC2 t3.large** instances
- Handled **80L+ transactions** by integrating **PhonePe API**, authentication using **JWT**, **OAuth2** and **rate-limited endpoints**
- Delivered **99.8% OTP/email delivery** using **MSG91**, **Mailer**, **async retry queues** and monitoring through **CloudWatch**
- Shipped **React Native app** on **Play Store (1K+ downloads)** with backend integration for **event access** and **admin tools**

Projects

KeystoneDB

Mar'25 - Apr'25

- Built a **lightweight RDBMS** in **C++** with an **SQL parser** using **Flex** and **Bison** and **persistent storage** by **RocksDB**
- Implemented **modular query execution** supporting **15+ SQL operations** over **LSM-tree** with **transaction handling**
- Developed a **custom key-value storage layer** using **RocksDB**, ensuring **durability** and **correctness** for SQL operations
- Added **70+ unit tests** with **Google Test** to verify **parsing accuracy**, **transaction consistency**, **safe concurrent execution**

Skills & Expertise

- Programming Languages:** C/C++, Rust, Go, Elixir, Python, TypeScript, JavaScript, Bash, Awk, Verilog, SQL
- Libraries and Frameworks:** React, Next.js, React Native, Node.js, Express, Pandas, NumPy, GraphQL, Prisma, Django, Flask
- Tools/Platforms:** Docker, Kubernetes, Redis, MongoDB, Postgres, AWS, Azure, Git, Databricks, Linux Kernel, Raspberry Pi

Relevant Courses

- Computer Science:** Programming and Data Structures, Algorithm-I, Software Engineering, Computer Organization and Architecture, Compilers, Database Management Systems, Operating Systems, Computer Networks
- Mathematics:** Advanced Calculus, Linear Algebra and Numerical and Complex Analysis, Probability & Statistics, Discrete Structures, Formal Language and Automata Theory
- More Courses:** Basic Electronics, Switching Circuits and Logic Design, Machine Learning, Data Analytics

Awards and Achievements

- Codeforces** - sumitkumar-17 **1700** (Expert) and **LeetCode** - SumitKumar-17 **2181** (Guardian)
- Secured **AIR 2038** in JEE Advanced and **1870** in JEE Mains (1.2M candidates) in 2022.
- Contributed to open-source orgs **NixOS**, **Stripe**, and **Supabase** by writing **patches**, refining **Nix expressions**, and improving **system builds**
- Served as the Unit Leader for NSS Unit-4, coordinated activities, and received the **Best Volunteer Award**